

Tajamul Khan

Gen AI Cheat Sheet

@Tajamulkhann

Types of Generative AI

TEXT GENERATION

- GPT-4, Gemini, Claude, Cohere
- Applications: creative writing, code generation, summarization, chatbots

IMAGE GENERATION

- DALL·E 3, Midjourney, Stable Diffusion, Firefly, StyleGAN, Google Imagen
- Applications: design, advertising, art, rapid prototyping

VIDEO GENERATION

- Sora (OpenAI), Synthesia, Google Veo, Runway
- Applications: marketing, animation, synthetic media, explainer content

CODE GENERATION

- GitHub Copilot, Gemini Code Assist, Claude, Code Llama
- Applications: productivity, automation, documentation, teaching

AUDIO GENERATION

- ElevenLabs, Suno, Murf, Udio
- Applications: voiceover, audiobooks, music

@Tajamulkhann

Data Preprocessing

Text

- Tokenization,
- cleaning (removing noise, correcting spelling),
- chunking for context windows

Images

- Normalization,
- resizing,
- aspect-ratio control

Audio

- Noise reduction,
- spectrogram prep

@Tajamulkhann

Leading Models and Architectures

Prompt Engineering:

- Use clear, descriptive, and specific prompts
- Use role instruction for the model (e.g., “Act as a science teacher”)

```
from transformers import AutoModelForCausalLM,
AutoTokenizer
model = AutoModelForCausalLM.from_pretrained('gpt-
4')
tokenizer = AutoTokenizer.from_pretrained('gpt-4')
prompt = "Explain quantum computing simply."
inputs = tokenizer(prompt, return_tensors="pt")
output = model.generate(**inputs, max_new_tokens=50)
```

@Tajamulkhann

How to Generate and Evaluate

Transformers (GPT-4, Gemini, Claude): backbone of all top LLMs and Vision-Language models

Diffusion Models (Stable Diffusion, DALL·E 3): state-of-the-art for images, art, and style transfer

GANs (StyleGAN, Pix2Pix): high-resolution faces, images, and creative art

Multimodal Models: Gemini, GPT-4o—combine text, image, and audio for universal understanding

@Tajamulkhann

Evaluation Metrics

Text: BLEU, ROUGE, Perplexity, Human-Rubric/Preference, Galileo, Vertex AI Rubrics

Image: FID, Inception Score, Human preference, style/consistency checks

Multimodal: Task-specific rubrics, real-world user feedback, adaptive rubrics

AI Safety: Hallucination checks, content moderation, factuality assessment

@Tajamulkhann

Feature Engineering & Fine-Tuning

- Use Embeddings for semantic search/zero-shot tasks (e.g., OpenAI embeddings, Cohere)

- Fine-tune foundation models on in-domain data for higher relevance

- Apply context window optimizations for long documents (chunking, retrieval-augmented generation)

@Tajamulkhann

Hyperparameter Tuning

- Batch size, learning rate, number of layers

- **LLMs:** Context length, Temperature, Top-p/nucleus sampling

- **Images:** Steps per image, noise level (for diffusion)

- Use tools like Vertex AI's evaluation and tuning, or Hugging Face Optuna

@Tajamulkhann

Top Generative AI Tools

Category	Top Tools/Models
Chat/Writing	GPT, Claude, Gemini
Images	Midjourney, DALLÂ·E 3
Video	Sora, Synthesia, Veo
Code	Copilot, Claude, Gemini
Audio/Music	ElevenLabs, Murf, Udio
Evaluation	Galileo, Vertex AI

@Tajamulkhann

Found Helpful ?

Repost



Follow for more!